

A Simple Method for Determining Cerebralization

BRAIN WEIGHT AND INTELLIGENCE

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ABSTRACT By means of comparing the weight of the brain with the weight of the spinal cord the authors seek a way to appraise the intelligence of the individual. Authors' concept, on the one hand, unites the advantages of the other methods, on the other hand, it reduces their errors. Namely, the weight of the spinal cord stands in proportion to the bulk of the body of the individual, its surface and its functions, while the brain—with the exception of a small somatic part standing in constant correlation with the spinal cord—serves as the seat of the intellect. In different vertebrates the weight of the brain surpasses that of the spinal cord in different degrees. The weight of the brain in the frog is twice as high as that of its spinal cord, in the dog six times, in the cercopithecus 14 times, in the orang-utan 33 times, and in man 43 times. Pertinent detailed data are revealed by the table.

I. The Problem

The determining of the degree of cerebralization and intelligence of the different species and groups of animals constitutes one of the fundamental and most interesting problems of zoology, comparative anatomy and biology. In many respects it also touches upon the various regions of practical life. From the viewpoint of the utilization of animals in the interest of man, the question of docility and stage of development play an important role. In fact, some authors even bring the age of animals into connection with their stage of development.

From a scientific and practical viewpoint it is therefore justified to raise the question as to how the stage of development of different animal species can be ascertained.

Numerous attempts have been made so far to solve the problems. All of them agree in endeavoring to establish some kind of connection between the problem and the nervous system of the animals.

II. Solutions attempted so far

The best known solutions are as follows:

Observation. In every day life an instinctive intelligence classification has come into being with regard to the conduct and docility of animals. For instance, the dog, horse, and monkey are considered

to be clever, brainy animals, while the hen and the cow are held to be rather stupid.

Franz ('35) introduces the concept of perfection, elevation. Anything that is differentiated and centralized is perfect. This is manifested in the increase of the number of individuals and groups, in the formation of systematically subordinate units and the elevation of the functional capacity (the functions—from the viewpoint of existence—become more and more productive and tend to run their course faster).

Sewertzoff ('31) speaks of progressive evolution. This implies, on the one hand, aromorphosis, the increase of life energy, and on the other hand, idioadaptation, i.e., the establishing of special and useful reactions in accordance with environmental alterations.

Though these observations contain numerous useful viewpoints, their greatest drawback is that they lack an objective, uniform basis for comparison.

Absolute weight of brain. As the bulk of the brain is considered to be the standard of the developmental stage of the nervous system within certain limits, we first of all compared the absolute brain-weight of various species.

The deficiency of the system soon came to light, for the elephant and the whale with their brains weighing 5 kg and 7 kg

respectively, exceed by far the weight of the most developed human brain (Lenhossék, '23; Lapicque, '44).

Relative brain-weight. It soon became apparent that the weight of the body must also be taken into consideration in making comparisons. True enough, the whale has a brain five times the weight of a man's, but the weight of its body is about one thousand times heavier than that of man. Thus the whale's 7 kg brain compared with its body is as small relatively as if a man had a brain the size of a hazelnut (Lenhossék, '23).

It was Cuvier who introduced the concept of the relative brain-weight. According to this

$$I = \frac{E}{C} \text{ gr in which}$$

I = degree of intelligence

E = brain-weight (encephalon)

C = body-weight (corpus)

A few relative brain-weights as calculated by Anthony ('28): hippopotamus: 1/2700, horse: 1/600, cat: 1/100, man: 1/45, mouse: 1/40, marmoset: 1/15.

Though the introduction of the relative brain-weight betokened progress, the method needed further perfection. It is quite obvious that the mouse is far surpassed by man as regards development, though it would precede man on the basis of its relative brain-weight, according to Cuvier.

Dubois (1897) pointed to the fact that the size of the brain is equal to the exponential function of the weight of the body.

$$E = KP^r \text{ in which}$$

E = brain-weight

K = cephalizational index,
developmental index

P = weight of body

r = relation exponent

Dubois composed his formula in an empirical way. He selected 2-2 animals neighboring on the assumption that their intelligence was about equal, and took the quotient of their brain-weights. Then he sought for the exponent:

$$\frac{\text{brain-weight } a}{\text{brain-weight } b} = \frac{\text{body-weight } a}{\text{body-weight } b} r$$

For the value of the r exponent he got 0.54-0.58, of which the mean value was

0.56. In possession of these exponents he calculated the developmental index (cephalization) of the brain of an animal with the following formula:

$$\text{Cephalization} = \frac{\text{brain-weight}}{\text{body-weight}^{0.56}}$$

This original concept of Dubois has been modified by numerous authors. In general they tried to determine the value of the relation-exponent and the cephalization index more precisely.

Brummelkamp ('39) altered the value of K. According to Dubois, K changes regularly: 1, 2, 4, 8 and 1, 1.5, 2, 3 . . . etc. According to Brummelkamp: $\sqrt{2}, \sqrt{2}^2, \sqrt{2}^3, \dots$. Thus $E = (\sqrt{2})^a / P^{5/9}$.

Wirz ('50) calls attention to the fact that the relation exponent determined by Linder ('46) by means of the linear regression method is not a constant number, the $r = 0.56$ given by Dubois is but an approximate limit value, so that the value of K as determined by Dubois and Brummelkamp would also be incorrect.

Lapicque ('24, '44) logarithmized the formula of Dubois: $\log E = \log K + r \log P$

The equation resembles the equation of the straight line: $y = ax + b$, represented by a coordinate system, in which the value of the ordinate is given by $\log E$, the abscissas by the value of $\log P$. It really gives a straight line, the direction-angle of which depends on the "r" relation exponent, its height on the value of "K." The equations of the animals of different intelligence yield straight lines that are parallel to one another. Dubois himself distinguished ten altitude degrees, each above the other. The first is the straight line of man with a cephalization index of 270.

The concept of Dubois, one of the fundamental methods, is used even today for comparing the various species of animals. This method, however, has its deficiencies:

a. If the body-weight is taken as basis, confusion is caused by the circumstance that the fat-contents may give rise to enormous weight differences which may cause great calculational deviations, for the fat contents does not stand in relation to the development of the nervous system. For this reason, some have tried to sub-

stitute the whole weight of the body with certain organs (thigh-bone, liver, etc.). No satisfactory result has been obtained so far.

b. The great failing of the method is that it rests on an empirical formula gained by approximate estimation.

c. The cephalization ranking disclosed by Dubois reveals certain unevennesses. It is highly improbable that the Indian elephant possesses a brain development (1.24) twice that of a *Hylobates lenciscus* (0.69) belonging to the ape family. The camel cannot be supposed to claim a higher place than the marmoset. Nor will the hippopotamus be two and a half times as intelligent as the rat. Manouvrier pointed out in 1885 already that two separate parts are to be distinguished in the brain from a functional viewpoint: a somatic part which is proportional to the bulk of the body, its surface and function (E_s), and a mental part standing in the service of the mind (E_a).

$$E = E_s + E_a$$

The somatic brain is dependent on the body-weight. $E_s = f/P = K' P^r$. Thus $E = E_s + E_a = E_a + K' P^r$. The Dubois formula $E = K P^r$ may therefore be modified as follows: $E_a + K' = K P^r$. From this:

$$K = \frac{E_a + K' P^r}{P^r} = \frac{E_a}{P^r} + K'$$

Manouvrier's method, though more perfect than the basal Dubois formula, is not free from failings either. As Lapique ('44) points out the $E = E_a + K' P^r$ formula resembles the equation of the straight line. Thus, related animals of identical intelligence should occupy places on the same straight line in the co-ordinate system. The parameters, however, yield a curved line, so that the hypothesis is not quite correct. As the body-weight appears as an independent variable here, the difficulties experienced within the Dubois formula are to be found here too.

The most highly developed, and, beside the formula of Dubois, the most frequently used method of solving the problem is the utilization of the intracerebral indices (Wirz, '50; Portmann, '48).

In the case of the simple, immanent indices, the mass of the lower vegetative

functional centers (brain stem) of the animal are compared with the mass of the higher integrating centers. The higher the quotient, i.e., the greater the bulk of the higher centers is, the greater the stage of development is.

A much more exact and unified picture can be obtained by employing the transcendental indices. In this case the parts of the brain of the animal in question are compared with the brain stem of another animal of the lowest development, assumedly of the same body-weight and belonging to the same group.

The most ancient form of the vertebrate class is represented by the insectivores (e.g., hedgehog or porcupine). Their brain stem is the least developed and may be represented by a straight line on which each size of body possesses a brain stem value. The brain stem value, which the animal in question would possess in proportion to its body-weight if it belonged to the insectivores, is called the basic number. Animals of similar size possess similar basic numbers. E.g., man's brain stem index, basic number is ten. This means that the brain stem of a man is ten times as large as the brain stem of a porcupine of similar size.

The individual indices are gained by dividing the brain part in question with the basic number:

$$\text{Complete brain index (T.i.)} = \frac{\text{complete brain}}{\text{basic number}}$$

$$\text{Neopallium index (N.i.)} = \frac{\text{neopallium}}{\text{basic number}} \dots \text{etc.}$$

The developmental rank of the animals may be arranged on the basis of the T.i. which is corrected in a small degree by the N.i. The bigger the N.i. is, and the smaller the brain stem index is expressed in T.i.%, the more developed the animal is. Table 1 shows a few examples according to Wirz ('50).

The solution of the problem, no doubt, may be approached most accurately by employing the intracerebral indices. This method, however, has its own failings, too.

a. The isolation of the brain stem is a very difficult and questionable operation in practice. If we wish to separate the cerebral centers of the somatic and mental

TABLE 1
Intracerebral indices according to Wirz

Vertebrate	Brain-stem index	Complete brain index	Neopallium index	N.i. T.i.	A.i. T.i.
	A.i.	T.i.	N.i.	%	%
Porcupine	1	3	0.78	26	29.9
Elephant	9.95	150	70	—	—
Macacus	4.07	—	32.4	—	—
Primates	—	—	—	78.3	9.2
Man	10	214	170	80	44.6

functions precisely, we come across obstacles which render it impossible for us to separate the brain physically into two parts, through all classes of vertebrates.

b. The relationship of the parts of the brain to the somatic and mental poles undergoes a change during the course of phylogenesis.

c. The calculation of the basic number is performed on the basis of the Dubois formula and thus bears its failings too.

III Concepts of the authors

Definition. The stage of development and intelligence of the vertebrates (I) is in direct proportion to the modification of the quotient of the brain-weight (E) (encephalon) and the weight of the spinal cord (Ms) (medulla spinalis) in the same animal.

$$I = \frac{E}{Ms}$$

Proof. The weight of the brain and the spinal cord have been compared by numerous authors, but no conclusions have as yet been deduced. Soemmering (1785) and Ziehen (1899) have ventured the remark that the weight of man's spinal cord, compared with his brain, is the smallest of all animals.

Our method appears to succeed in summing up the advantages of the procedures up to now, and eliminate all their greater failings.

Its advantage, compared with the concepts of Dubois and others' similar to his, is that it puts an end to the mistakes resulting from the utilization of the weight of the body which displays a broad scale of variants independent of the stage of development (tall — small, stout — lean). The spinal cord, on the other hand, represents the whole of the body-weight cor-

responding to the actual developmental and differentiation degree.

Lapicque ('44) sets up exact interrelations between the cross-section of the spinal cord and the number of fibers, the cross-section of the fibers and the weight of the body. Latimer ('50) finds — on the basis of measurements carried out on guinea pigs — that the spinal cord is in much closer connection with the dimensions, length, and weight of the body than the brain, which is, to a certain degree, independent of the weight of the body.

Besides this, the spinal cord, by means of its development and structure, is far more suitable for comparison to the brain than the weight of the body. In this way, namely, we compare common denominator parts.

Our method is most similar to that of the intracerebral indices. While Wirz compares the brain and the weight of the individual parts of the brain with the brain stem, we consider the basis of comparison to be the spinal cord. The two parts of the central nervous system are namely the somatic and mental parts. The proportion of the two would give us most clearly and comprehensively a measure of the degree of intelligence. The purely somatic part is the spinal cord, the mental part is represented by the cortex, or its corresponding structures. The medial parts — thus the brain stem, too — reveal a mixture in the relations of the two functions.

There is no doubt that not only the spinal cord, but also the somatic centers of the brain change as a consequence of the alteration taking place in the somatic part, thus the value of the quotient undergoes a certain distortion. This is the most important weakness of our procedure. At

the same time it is indisputable that the change of the somatic centers in the brain, on account of the ever-increasing mental functions, is of considerably smaller extent than the change of the purely somatic spinal cord. We get a truer picture by all means, if in the place of the brain

stem we put the spinal cord in the denominator of the fraction. The proportion shift—amounting to a small percent—which may be expected on the basis of the anatomical structure does not show an essential deviation even in the animals of the lowest order.

TABLE 2
The brain- spinal cord-weight proportion of vertebrates

Name	Author	Body-weight (C)	Brain-weight (E)	Spinal cord-weight (Ms)	E Ms
		<i>g</i>	<i>g</i>	<i>g</i>	
Homo sapiens	Krompecher	54,333	1,273.7	29.7	42.8
Orang-utan	Krompecher	21,000	316.0	9.5	33.3
Seal (<i>Phocoena communis</i>)	Ziehen	—	—	—	20.0
Blackheaded parrot (<i>Adapornis personata</i>)	Krompecher	40	1.92	0.12	16.0
Cerkof monkey (<i>Macacus sinicus</i>)	Krompecher	1,538	67.5	4.7	14.3
Sparrow (<i>Passer domesticus</i>)	Krompecher	27	1.0	0.085	11.8
Greyback crow (<i>Corvus cornix</i>)	Ziehen	—	—	—	11.4
Finch (<i>Fringilla domestica</i>)	Ziehen	—	—	—	10.9
Swallow (<i>Hirundo rustica</i>)	Krompecher	18	0.535	0.08	6.7
Swallow (<i>Hirundo rustica</i>)	Krompecher	20	0.492	0.075	6.6
Dog (<i>Canis familiaris</i>)	Krompecher	7,150	62.0	10.0	6.2
Dog (<i>Canis familiaris</i>)	Krompecher	12,300	83.0	13.5	6.1
White mouse (<i>mus musculus albus</i>)	Krompecher	16.5	0.425	0.075	5.6
Bat (<i>Vespertilio serotinus</i>)	Ziehen	—	—	—	5.1
White mouse (<i>mus musculus albus</i>)	Krompecher	19.6	0.43	0.085	5.0
Shrew-mouse (<i>Sorex-vulgaris</i>)	Ziehen	—	—	—	4.9
Grey rat (<i>Mus decomanus gris</i>)	Krompecher	77	1.66	0.35	4.7
Mole (<i>Talpa europaea</i>)	Ziehen	—	—	—	4.7
Domestic mouse (<i>Mus musculus</i>)	Krompecher	—	0.35	0.075	4.6
Grey rat (<i>Mus decomanus gris</i>)	Krompecher	85	1.75	0.4	4.4
Squirrel (<i>Sciurus vulgaris</i>)	Ziehen	—	—	—	4.3
White rat (<i>Mus decom. alb.</i>)	Krompecher	213.5	1.93	0.53	3.6
Guinea-pig (<i>Cavia porcellus</i>)	Latimer	649.4	4.0	1.1	3.6
Porcupine (<i>Erinaceus europaeus</i>)	Ziehen	—	—	—	3.5
Sheep (<i>Ovis aries</i>)	Krompecher	—	102.0	30.0	3.4
Cat (<i>Felis domestica</i>)	Krompecher	4,650	32.0	9.5	3.3
White rat (<i>Mus decom. alb.</i>)	Krompecher	211	1.793	0.547	3.3
Pigeon (<i>Columba domest.</i>)	Krompecher	335	2.03	0.62	3.2
Marmot (<i>Arctomys marmota</i>)	Krompecher	1,980	13.2	4.15	3.14
Pigeon (<i>Columba domest.</i>)	Krompecher	326	1.79	0.575	3.1
Gopher (<i>Cricetus frument.</i>)	Ziehen	—	—	—	3.1
Horse (<i>Equus caballus</i>)	Krompecher	—	532.0	215.0	2.5
Rabbit (Race X) (<i>Oryctolagus cuniculus</i>)	Latimer	1,565.4	9.1	3.9	2.3
	Latimer	1,734.9	0.3	4.2	2.2
Hen (<i>Gallus domest.</i>)	Krompecher	1,340	3.525	1.554	2.2
Frog (<i>Rana esculenta</i>)	Krompecher	48	0.117	0.054	2.1
Rabbit (Race III) (<i>Oryctolagus cuniculus</i>)	Latimer	2,661.2	11.7	5.7	2.0
	Latimer	2,498.2	11.5	5.9	1.9
Cock (<i>Gallus domest.</i>)	Krompecher	1,800	3.68	2.175	1.7

C, corpus; E, encephalon; Ms, medulla spinalis.

Finally, from a technical viewpoint, it is far easier and much more exact to separate the spinal cord than to isolate the brain stem and other parts of the brain.

METHOD

The brain and the spinal cord were weighed without fixation in a fresh state after removal, without the dura mater. The pia mater, however, was kept. The roots of the spinal cord were cut off quite close to the cord, so that the whole of the cauda equina was removed.

RESULTS

Besides our own measurements (Krompecher, '33) we utilized the pertinent data of other authors (Ziehen, 1899; Latimer, '50, '59) too. The results of our measurements are summarized in table 2. The graphic delineation of a few interesting data are shown in table 3.

TABLE 3
COMPARISON OF BRAIN AND SPINAL CORD WEIGHTS

SPECIES	RATIO	SPINAL CORD	BRAIN
MAN	1:43		
ORANG	1:33		
PARROT	1:16		
DOG	1:6		
CAT	1:3		
HEN	1:2		

DISCUSSION

The data referring to man were gained from the weight-relations of 24 individuals. To obtain the average shown by the table, we took into consideration the weight-relations of only the 12 individuals between the ages of 28 and 62 (7 men and 5 women). Taking into account all our measurements, the extreme values of the data of 11 men were 39.3 and 45.3. The latter ratio number was given by a 38 year old man as well as a 51-year-old one. The extreme values of the 11 women were 37.8, and 51.1. Measurements on undeveloped individuals always show a more favorable ratio. This finding is in harmony with our knowledge regarding the development of the brain. Thus the

ratio of two girls (15 and 19 years of age) was 1:49. After the full development of the organism, the weight-proportion between the brain and the spinal cord does not change essentially, even in the case of pathologic obesity or loss of weight. An emaciated 106-year-old woman weighing only 23.10 kg yielded a 1:38.5 weight relation of the spinal cord, a result in conformity with the normal range.

A survey of the results gained from our measurements reveals that man towers far above all the animals in a most distinguished position.

A valuable index number in our studies is that of an orang-utan. The examined animal was a young one. Anthony gives an account of the body-weight of two adult male orangs: 73.500 and 76.500 g. Their brain-weights were 400 g and 395 g, respectively. It may be assumed that the brain—spinal cord-weight proportion of the adult animal is lower than the high number of 33. The seal occupies a conspicuously high place. The brain-weight relation of the aquatic animals yielded high values with all the methods used until now. Legendre ('12) attained this result by comparing the weight of the brain to that of the body, and endeavored to explain this finding with the thickness of the myelin sheath of the nerve fibers. This supposition proves untenable on the ground of comparison with the spinal cord, as the nerve fibers of the myelin sheaths in the spinal cord are likewise thicker. For explanation we must call attention to the amazing skill and docility of the seals in their circus feats, bearing in mind on the one hand, that their quite astonishing performances are carried out on dry land, which is not their element and where they can move about only with difficulty; on the other hand, that man appraises the intelligence of an animal mostly on the ground of his individual impressions. This appraisal may be, at the most, only informative, but not of decisive or classifying value. In the case of the aquatic animals we must mention those peculiar anatomical conditions which arise with regard to the reduction of extremities in consequence of adaptation to life in the water. In conformity with the rudimentary extremities, the surface of the body has

decreased, and, as result of this, the cervical and lumbosacral enlargements of the spinal cord (which comprises a rather considerable part of it) are lacking, or at any rate reduced. As a consequence, the spinal cord of the aquatic mammals becomes somewhat smaller, and, in proportion to this, the weight of the brain somewhat heavier.

The fishes are affected even more in this respect than the aquatic animals, but even so they occupy a low position. However, this calculating system may be employed without reservation when comparing them to one another.

The possibility of comparison without reservation likewise applies to the comparison of similar aquatic mammals with one another.

We must also mention the values pertinent to the mice and swallows. If we compare the brain-weight of these small animals to their body-weight, we find that the brain-weight of the mouse is just $1/38$ of its body-weight, that of the male swallow comprises but $1/33$ of its body-weight, so that both animals have a larger brain in comparison to their bodies than man $1/45$. If however, we employ the above mentioned spinal cord comparison method in calculating the relative weight of the brain, we find the place of the mouse near that of the rat with a relative number of 5.5, while that of the swallow would be in the middle of the bird scale with a value of 6.5.

The brain-spinal cord-weight proportion of young animals shifts along the scale in favor of the brain. The brain, being a highly differentiated organ — as we know from embryology — develops rather early. For this reason, young animals must be omitted from the comparative list, as is required in the method of Manouvrier and Dubois.

It will require investigation of a more comprehensive material to set up an intelligence scale within the individual classes of the animal world and compare it with individuals of other classes. On the basis of the few data at our disposal we are able to determine only a few members of this table. As regards the birds, for example, it may be ascertained that they are found within the values 2–16

(see table 2) according to our data gained so far. The question arose as to how we could explain that the dove occupies a higher stage than the horse. The explanation lies in the fact that the dove possesses an excellent faculty of direction. This faculty obviously is not without its suitable cerebral center.

It would be incorrect to raise the question, for example, whether the mammals are more intelligent than the birds, because a glance at the table would reveal that beside birds of extremely low intelligence we can find such as the magpie or the parrot whose stage of brain-development surpasses that of many mammals, whereas the hen is on the same level as the frog.

In this paper we desired to deal only with the weight proportions of the brains of the vertebrates on a comparative basis. According to the general conception, in addition to the weight deviation, the density and quality, as well as the depth of the gyrification of the brain surface is also of great significance. From the viewpoint of intellectual faculty it is obvious that the number of cells in the cerebral cortex, their connections and their quality are decisive factors. The gyrification of the surface of the brain is of significance primarily in comparing individuals closely related. Our aim is not to discuss the subtle differences between human beings or species of animals, but to classify creatures that stand quite distant from each other on a common denominational basis.

The data enumerated here must be considered as informative in character. Much more data are needed, however, not only as regards the different species, but also concerning the individual within certain species. The classification disclosed here may prove to be helpful with regard to the further development of the method, furthermore, to aid experts exercising due caution to appraise the cerebralization of animals and thereby to gain information as to whether it would be worth while to occupy oneself with an animal from the viewpoint of docility.

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